Agenda

2. Convolutional Neural Networks (CNNs)

2.1 From MLP to CNN

2.2 Core Components of CNNs

- Convolution ✓
- Pooling
- Fully Connected Layers

2.3 Full CNN Pipeline

2.4 Training CNNs

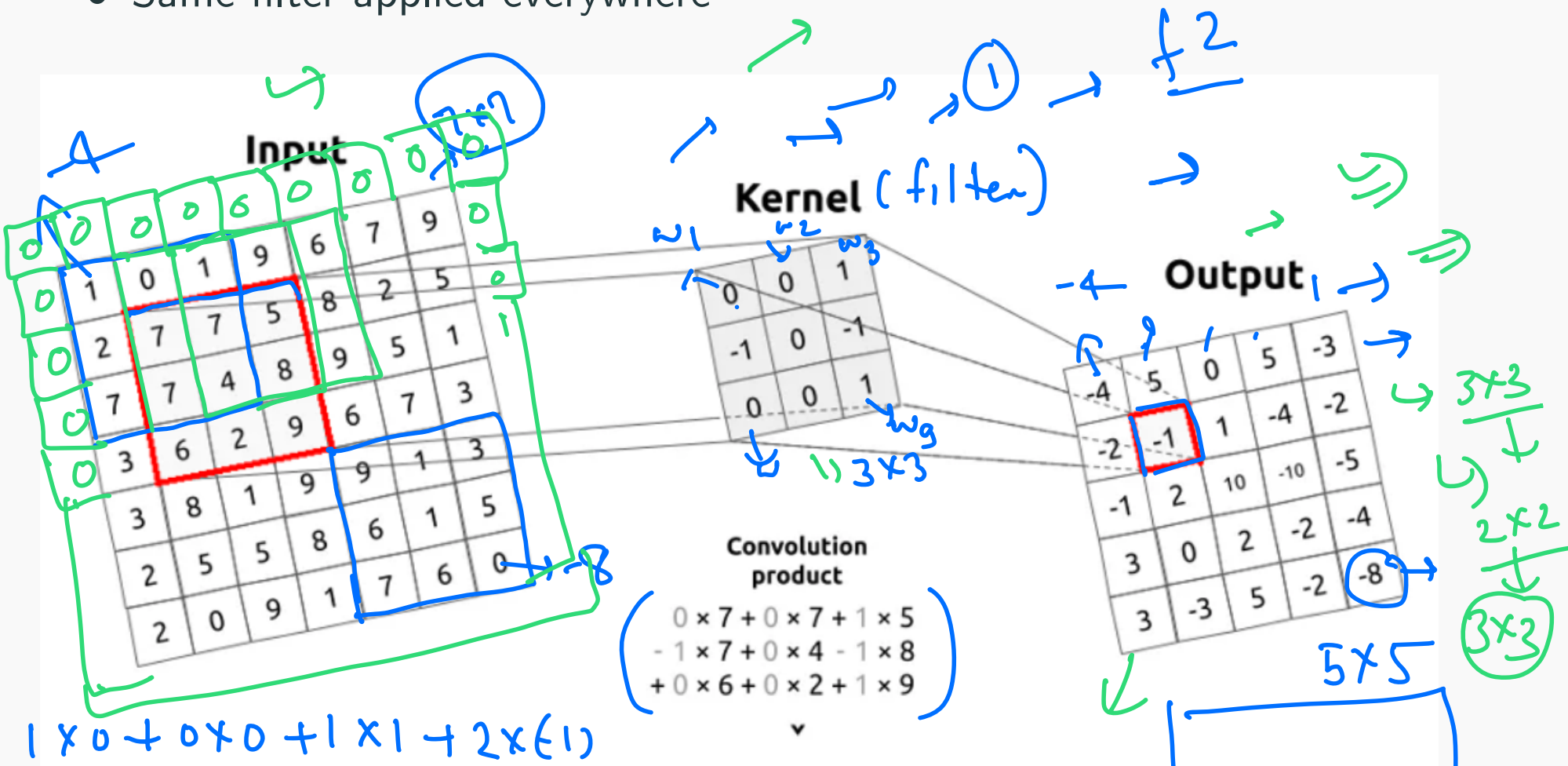
2.5 Practical Tips

2.6 Popular CNN Architectures

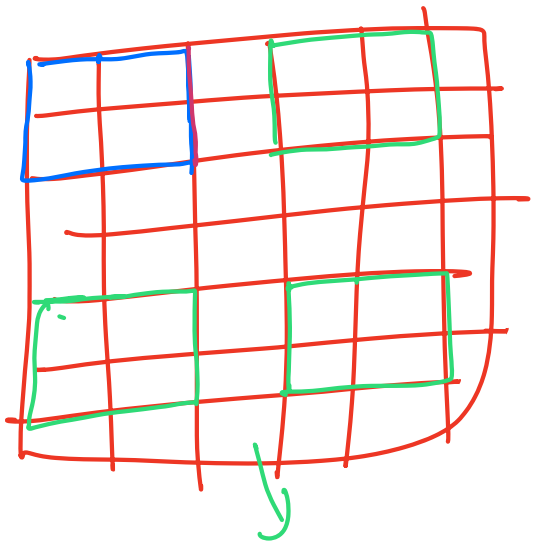
2.7 CNNs Lab

What is Convolution? (Intuition)

- A small filter scans across the image
- Detects patterns like edges, textures
- Same filter applied everywhere

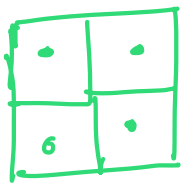


$1\times0+0\times0+1\times1+2\times(-1)$
$+7\times0+(-1)\times7+\ldots=-4$



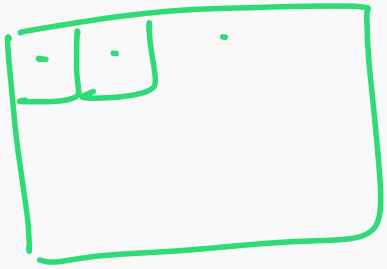
S=3

2×2



Convolution Operation

$$Y[i,j] = \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} W[u,v]\, X[i+u, j+v]$$



- W: filter (kernel), learned during training
- X: input image (or previous feature map)
- Y: output feature map (one map per filter)
- Intuition: each output pixel is a *weighted sum* of a local patch

Learn more from: https://developer.nvidia.com/discover/convolution

Channels (Depth)

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$$X \in \mathbb{R}^{d_1 \times d_2 \times \cdots \times d_k \times C}$$

where C denotes the number of channels, and each channel is:

$$X[:, :, \ldots, :, c]$$

GrayScal $\Rightarrow X \in R^{H \times W \times 1}$

RGB $\Rightarrow X \in R^{H \times W \times 3}$

$X \in R^{H \times W \times C}$

$X \in R^{D \times H \times W \times 3}$

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kernel

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- **Interpretation:** A channel captures one type of information or feature across the spatial/temporal dimensions.
- **Special cases:**
  - **1D data (signals, time series):** $X \in \mathbb{R}^{L \times C}$
    Channel = 1D signal
  - **2D data (images):** $X \in \mathbb{R}^{H \times W \times C}$
    Channel = 2D map (e.g., RGB planes)
  - **3D data (videos, volumetric):** $X \in \mathbb{R}^{D \times H \times W \times C}$
    Channel = 3D volume

Filters (Kernels): How Channels are Produced i

- A **filter/kernel** detects local patterns by sliding over the input
- **Key idea:**
  - Each filter spans **all input channels**
  - **One filter** $\Rightarrow$ **one output channel**
  - K **filters** $\Rightarrow$ K **output channels**
- **1D Convolution (Time Series)**
  - Input: $X \in \mathbb{R}^{L \times C}$
  - One filter: $W^{(m)} \in \mathbb{R}^{k \times C}$
  - Output (one channel):

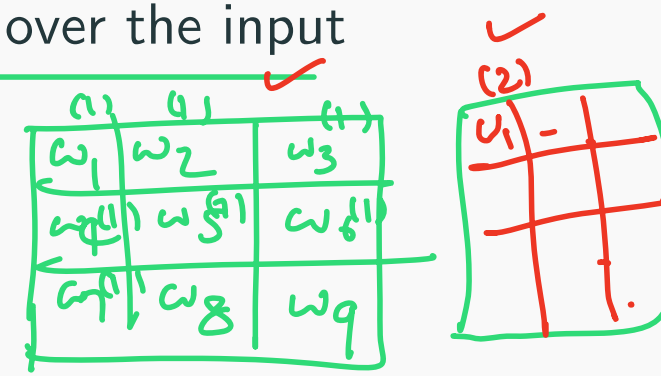
m=1

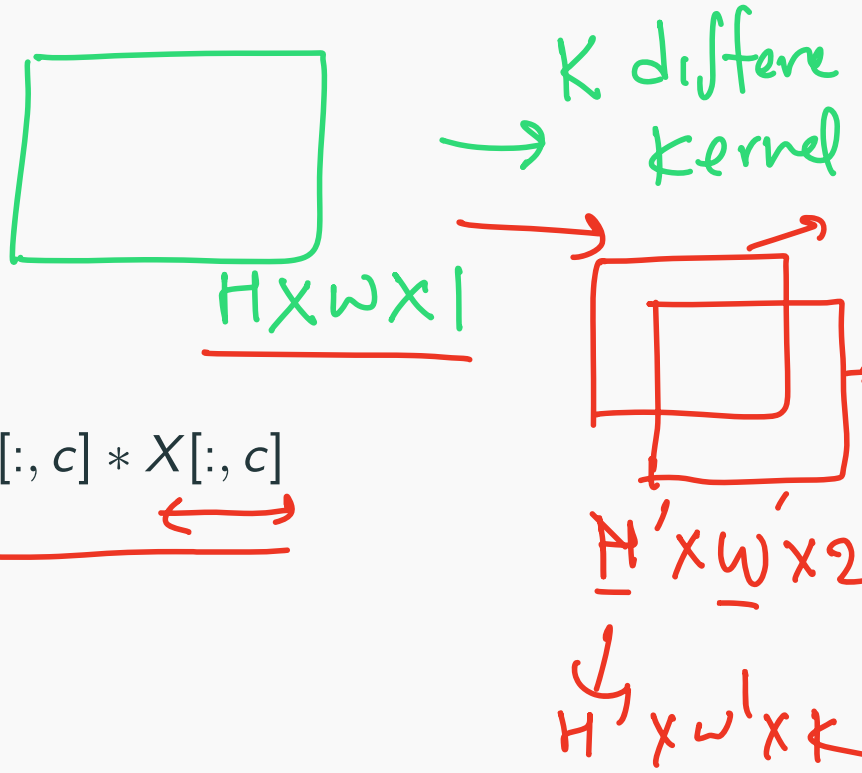
$$Y[:, m] = \sum_{c=1}^{C} W^{(m)}[:, c] * X[:, c]$$

m=1..K

  - Using K filters $\Rightarrow$ $Y \in \mathbb{R}^{L' \times K}$.

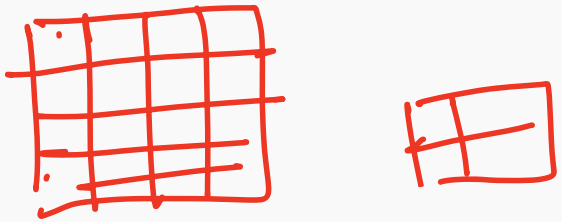
$L' < L$





Filters (Kernels): How Channels are Produced ii

- **2D Convolution (Images)** RG
  - Input: $X \in \mathbb{R}^{H \times W \times C}$
  - One filter: $W^{(m)} \in \mathbb{R}^{k \times k \times C}$
  - Output (one channel):



$$\boxed{Y[:,:,m] = \sum_{c=1}^{C} W^{(m)}[:,:,c] * X[:,:,c]}$$

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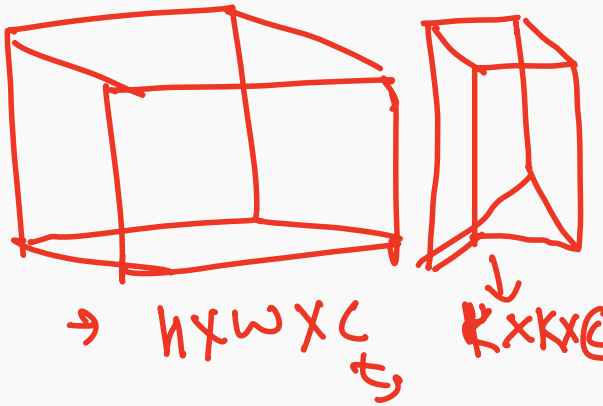


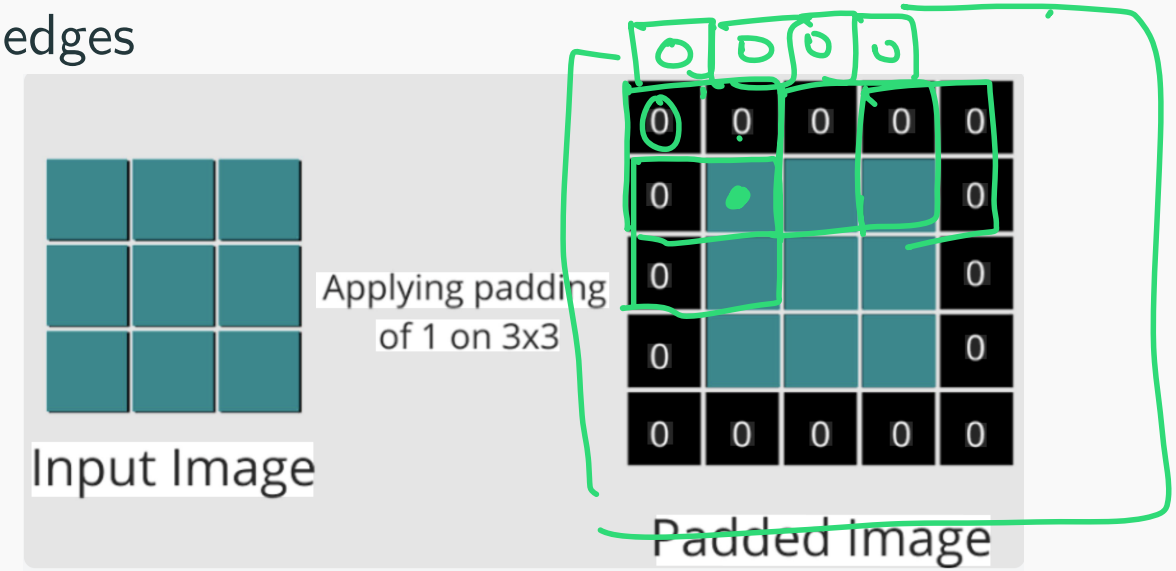
- **3D Convolution (Videos / Volumes)**
  - Input: $X \in \mathbb{R}^{D \times H \times W \times C}$
  - One filter: $W^{(m)} \in \mathbb{R}^{k \times k \times k \times C}$
  - Output (one channel):

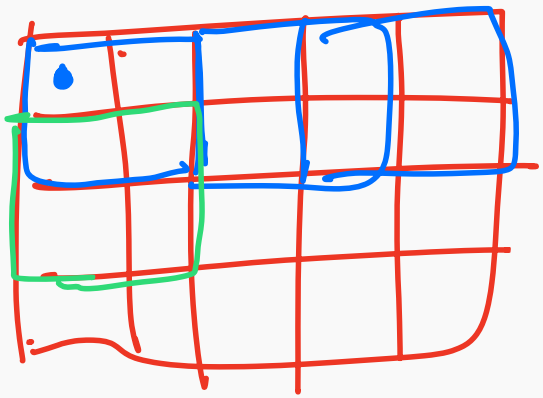
$$Y[:,:,:,m] = \sum_{c=1}^{C} W^{(m)}[:,:,:,c] * X[:,:,:,c]$$

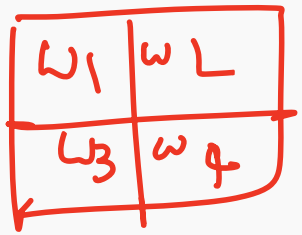
  - Using K filters $\Rightarrow Y \in \mathbb{R}^{D' \times H' \times W' \times K}$

Padding in CNNs: When and Why?

- **Padding** adds border pixels (usually zeros) so convolution can be applied near edges







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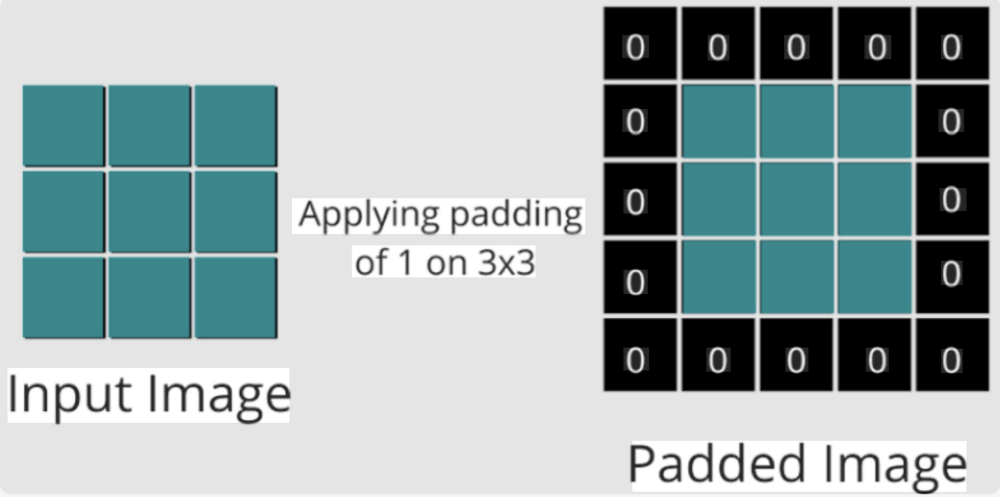

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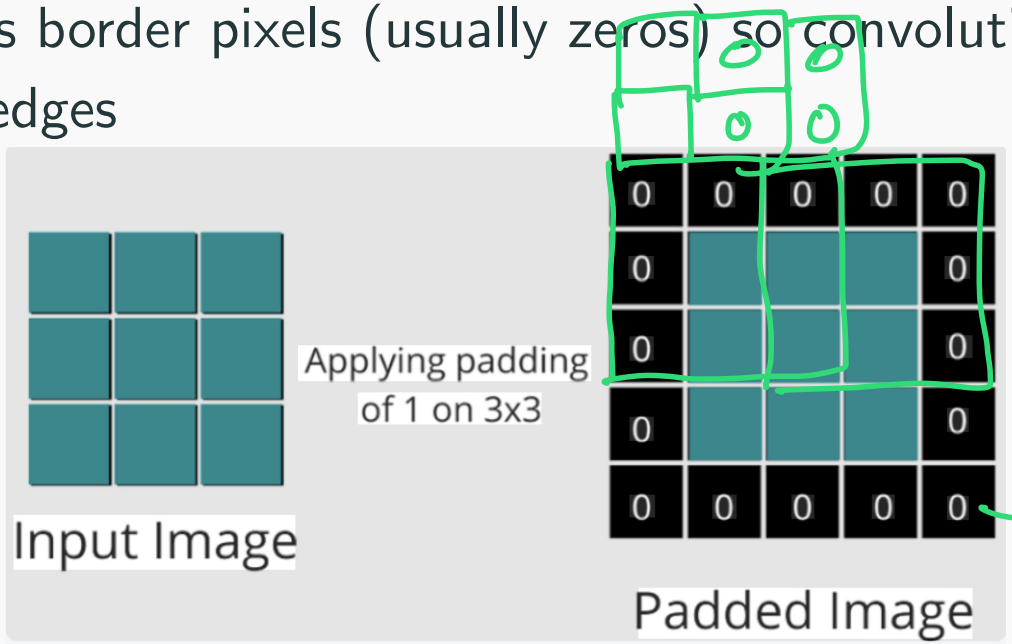

3x3 ⇒ s=2 2x2

5x5 → 5x5

- Why it matters:
  - avoids shrinking the feature map after every conv layer
  - preserves information at image borders (otherwise borders are seen fewer times)
  - helps keep shapes aligned across layers (useful in deep CNNs / skip connections)
- Two types of padding:
  - **Valid** padding: $P = 0$ (no padding) $\Rightarrow$ output shrinks
  - **Same** padding: choose P to roughly preserve size

Activation Functions in CNNs: Why and When?

- **Why do we need them?**
  - Without activation functions, the network is just a **linear model**
  - Stacking linear layers $\Rightarrow$ still linear (no extra power)
  - Activations introduce **non-linearity** $\Rightarrow$ can learn complex patterns

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$$\mathrm{ReLU}(x) = \max(0, x) = \begin{cases} 0 & x < 0 \\ x & x > 0 \end{cases}$$

$\Rightarrow \sigma(w_1 x_1 + w_2 x_2 + \dots + w_9 x_9)$

  - Very simple and computationally efficient
  - Avoids vanishing gradients (unlike sigmoid/tanh)
- **Leaky ReLU**

$$\mathrm{LeakyReLU}(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}, \qquad \alpha > 0 \quad \text{is a small constant}$$

$\alpha = 0.01$

  - Fixes "dead neuron" problem in ReLU
  - Allows small gradient when $x < 0$
- **When are they applied?**
  - Applied **after every convolution layer**: Typically `Conv` $\rightarrow$ `ReLU`

Batch Normalization: Why and How?

- **Why do we need BatchNorm?**
  - Deep networks are hard to train due to:
    - Changing distributions of activations (internal covariate shift)
    - Exploding / vanishing gradients
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- **What does it do?**

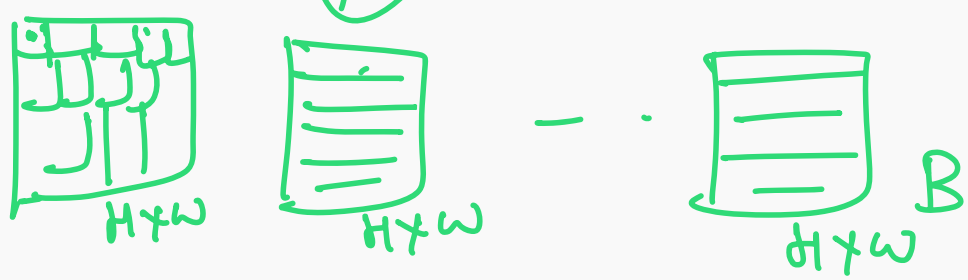
$$\Rightarrow \hat{x}_{ijb} = \frac{x_{ijb} - \mu_{ijb}}{\sqrt{\sigma_{ijb}^2 + \varepsilon}}, \quad i = 1 \ldots H, \quad j = 1 \ldots W$$

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y = \gamma \hat{x} + \beta$$

  - Normalizes activations using mini-batch statistics
  - Then rescales using learnable parameters (γ, β)

$$\mu_B = \frac{1}{b} \sum_{i=1}^{b} x_i$$

$$\sigma_B^2 = \frac{1}{b} \sum_{i=1}^{b} (x_i - \mu_B)^2$$



x_{ijb}

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$$\sigma_{ijB}^2 = \frac{1}{B} \sum_{b=1}^{B} (x_{ijb} - \mu_{ijB})^2 \rightarrow i = 1 \ldots H, \; j = 1 \ldots W$$

$B = 10$

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- **Where is it applied?**
  - Typically: `Conv` $\rightarrow$ `BatchNorm` $\rightarrow$ `ReLU`

Key Hyperparameters in Convolution

parameters

- **Kernel size** ($k_h \times k_w$): size of filter window (e.g., 3×3)
- **Stride** (S): step size when sliding the filter (e.g., $S = 2$ downsamples)
- **Padding** (P): zeros added at borders to control output size
  - **Valid**: no padding ($P = 0$)
  - **Same**: output size same as input
- **Number of filters** (K):
  - Determines number of output channels
- **(Optional) Dilation**:
  - Expands receptive field without increasing parameters

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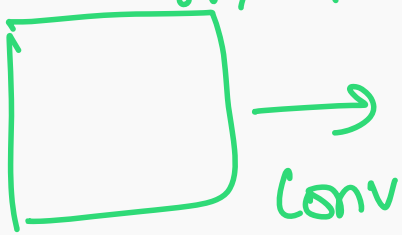
input $H \times W \times C$



conv

$H' \times W' \times K$

Output Size Formula

$$H' = \left\lfloor \frac{H + 2P - k_h}{S} \right\rfloor + 1, \qquad W' = \left\lfloor \frac{W + 2P - k_w}{S} \right\rfloor + 1$$

Output: $X' \in \mathbb{R}^{H' \times W' \times K}$

- Example: $H = W = 32$, $k = 3$, $S = 1$, $P = 1 \Rightarrow H' = W' = 32$ (same)

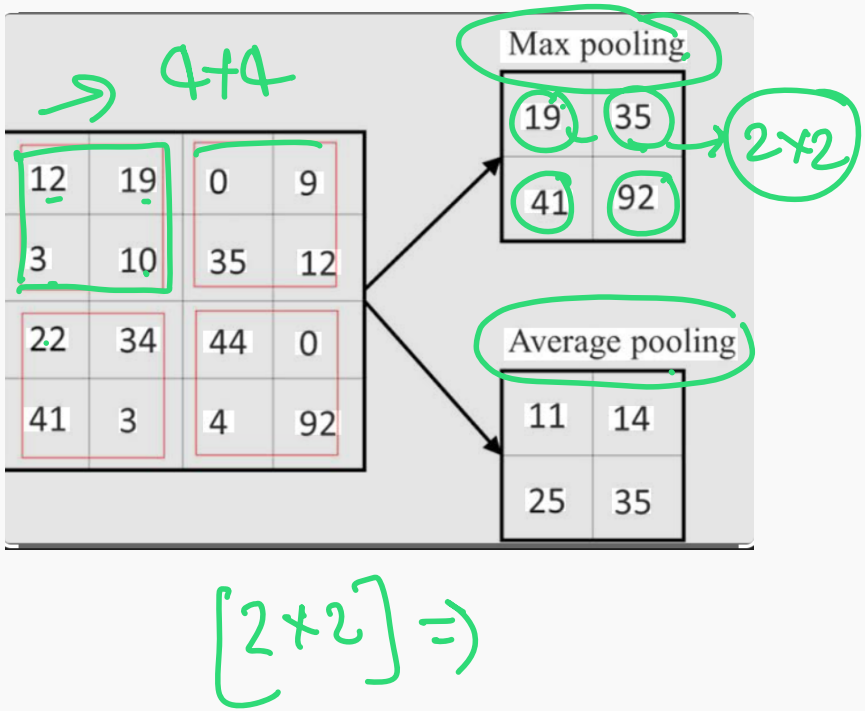
32×32 $\quad 3 \times 3$ $\quad P = 1$

Why Pooling?

- **Reduces spatial size**:
  - From $H \times W \to H' \times W'$ $\Rightarrow$ $H' << H$, $W' << W$
  - Reduces computation and memory
- **Provides translation invariance**:
  - Small shifts in input do not change output much
- **Increases receptive field**:
  - Each pooled value summarizes a larger region
- **Reduces overfitting**:
  - Acts as a form of downsampling / regularization

Pooling Types (and When to Use Them)

- **Max Pooling**
  - Takes the maximum value in each window
  - **Important when:** you care about *whether a feature exists* (edge/texture presence), and want robustness to small shifts
  - Common in early CNNs and in feature-extraction backbones
- **Average Pooling**
  - Takes the mean value in each window
  - **Important when:** you want *smoother* / less noisy activations (e.g., low-texture regions, denoising-like effects)



How Pooling Works

- Applied **independently on each channel**
- For input:

$$X \in \mathbb{R}^{H \times W \times C}$$

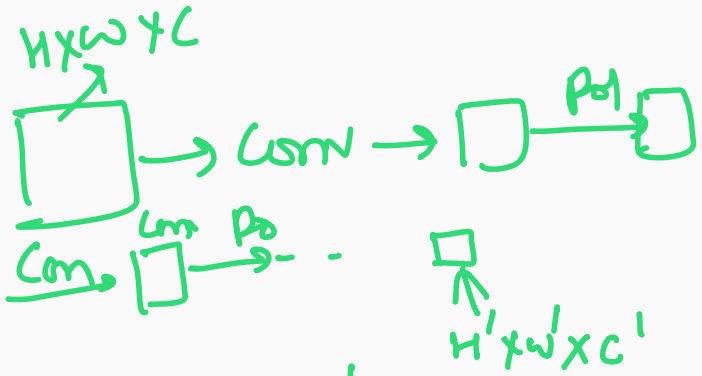
- A window of size $k \times k$ slides with stride s
- Output:

$$Y \in \mathbb{R}^{H' \times W' \times C}$$

- **Important:**
  - Number of channels C remains unchanged
  - Only spatial dimensions are reduced

Fully Connected (Dense) Layers i

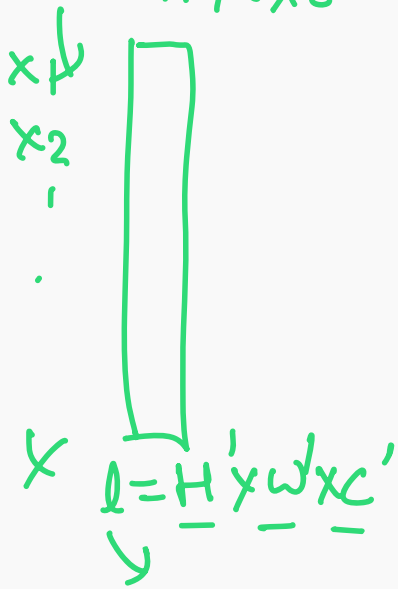
- A **Fully Connected (FC)** layer connects every input unit to every output unit
- **Role in CNNs:**
  - Placed near the **end of the network**
  - Combines all extracted features **globally**
  - Performs **final prediction / classification**



- **Input:**
  - Feature maps are **flattened** into a vector

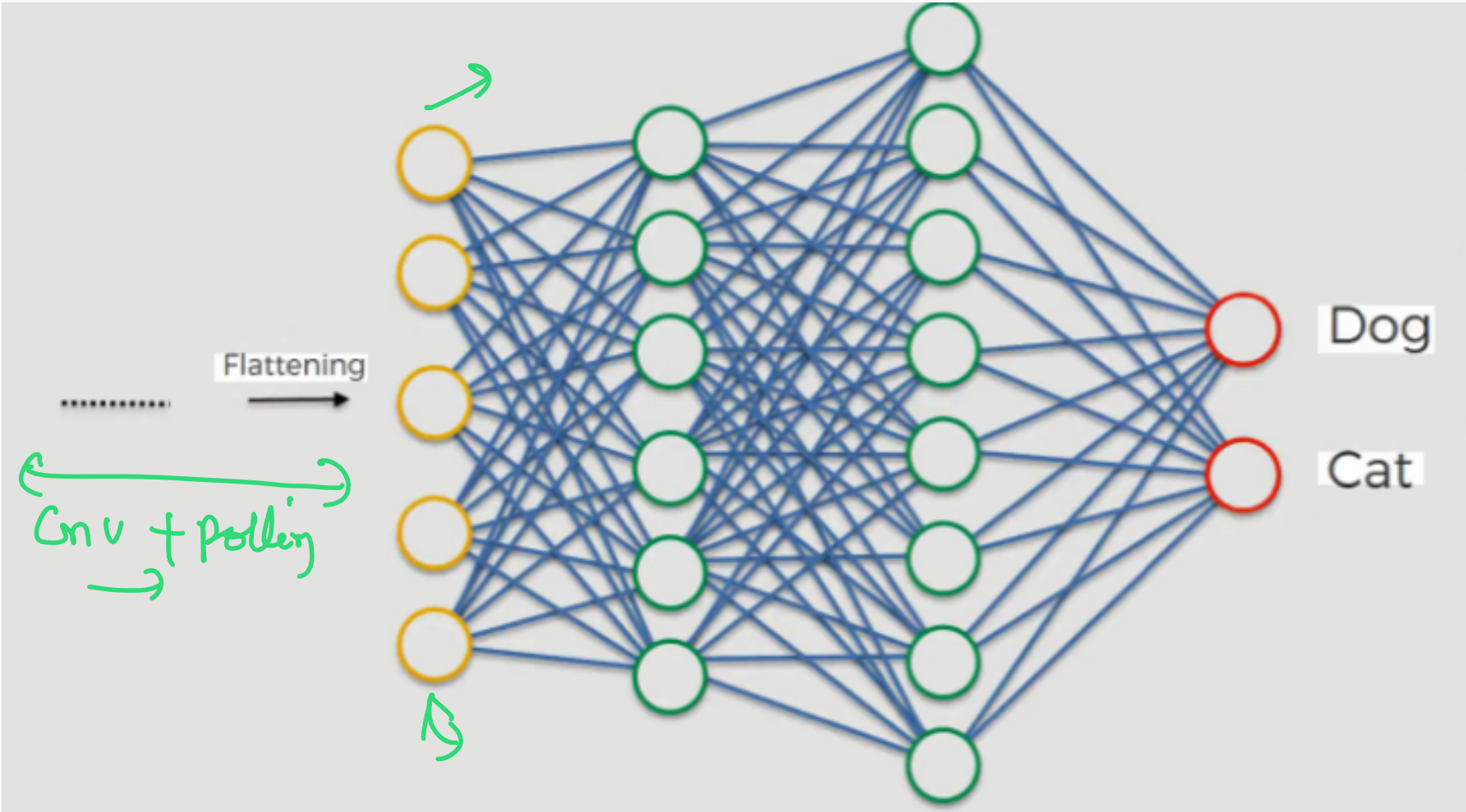
$$X \in \mathbb{R}^{H \times W \times C} \rightarrow \mathbf{x} \in \mathbb{R}^{n}$$

  where $n = HWC$



- **Interpretation:**
  - Learns **global combinations** of features
  - Acts like a standard **neural network layer**
  - Has **large number of parameters** (can cause overfitting)

Fully Connected Layers ii



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2.1 From MLP to CNN

2.2 Core Components of CNNs

Convolution

Pooling

Fully Connected Layers

2.3 Full CNN Pipeline

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2.6 Popular CNN Architectures

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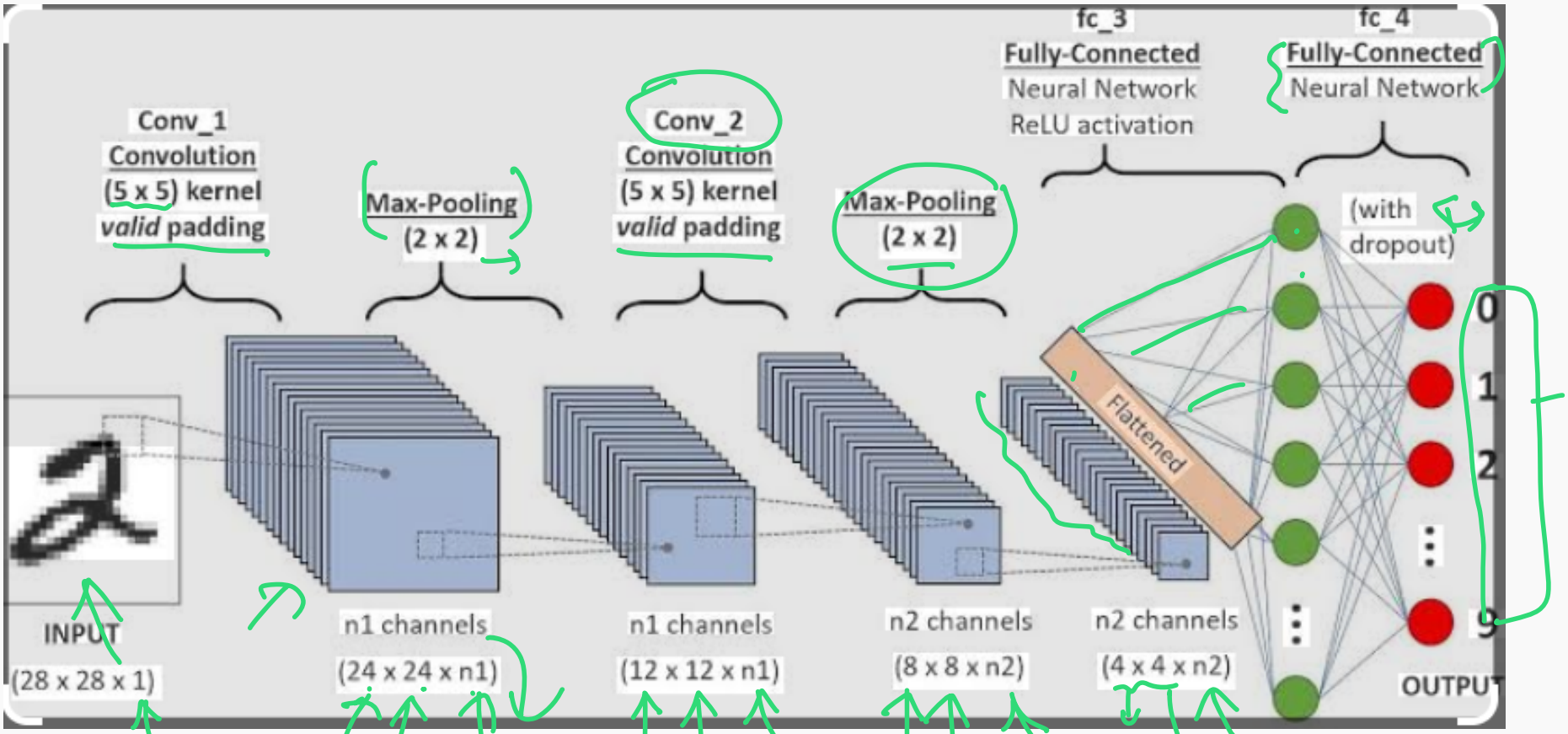
One Example of CNN Architecture

Input → Conv → ReLU → Pool
→ Conv → ReLU → Pool
→ Fully Connected → Softmax (for classification)



(28x28x1)

h', w

4x4xn2

→ 160

n2=10

n2=5